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Do Expressive Suppression and Cognitive Reappraisal Affect Memory?

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A widely cited idea in the emotion regulation literature holds that expressive suppression impairs memory while cognitive reappraisal has no effect on or may enhance memory relative to control. However, empirical evidence for these effects has been inconsistent. To provide a definitive test, we conducted four well-powered experiments with more than 4,000 participants to examine the effects of reappraisal and suppression of negative emotion (Experiments 1 and 3) and positive emotion (Experiments 2 and 4) on verbal and nonverbal memory. Results showed no consistent evidence for an effect of reappraisal on either type of memory, though expressive suppression of negative emotion consistently impaired verbal memory relative to control. Obtained effect sizes were small given successful emotion regulation manipulations and adequate statistical power ($ds < 0.11$). Conclusions are constrained to English-speaking online samples, and may not generalize to other types of memory. These findings highlight the need to systematically test widely accepted assumptions as a field.

Keywords: emotion regulation, expressive suppression, cognitive reappraisal, memory

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In addition to having affective and social consequences, emotion regulation is thought to impact cognition, including memory. A widely cited idea in the literature is that expressive suppression impairs memory while cognitive reappraisal does not affect memory or may improve it relative to control (Gross, 2013). Indeed, at the time of writing, a seminal article on this topic has been cited more than 2,500 times (Richards & Gross, 2000). Theoretical explanations for this idea hold that suppression may impair memory because it requires ongoing self-monitoring to inhibit behavioral expressions (Carver & Scheier, 1981), while reappraisal may not affect memory as it changes the meaning of an emotional stimulus early in the emotion generation process, rendering self-regulation unnecessary (Richards, 2004). An alternative theoretical perspective suggests that reappraisal may actually enhance memory as it requires deep elaboration of a stimulus in order to reduce its emotional impact (Dillon et al., 2007). Reconciling and testing these alternative perspectives, we went back to basics: conducting four well-powered experiments to comprehensively test the impact of suppression and reappraisal on verbal and nonverbal memory.

Emotion Regulation Strategies

The original process model of emotion regulation (Gross, 1998) argued that strategies to regulate emotion can be distinguished according to the temporal unfolding of an emotion generation process. According to this perspective, *antecedent-focused* strategies occur relatively early in emotion generation and aim to alter emotional response tendencies before they are fully activated. By contrast, *response-focused* strategies take place after emotion responses are generated and involve inhibiting an emotional response. Within these categories, cognitive reappraisal (antecedent-focused) and expressive suppression (response-focused) are among the most studied strategies as they tend to have contrasting affective, social, and cognitive outcomes (Gross, 2002, 2013; Sheppes, 2020).

Reappraisal involves changing the meaning of an emotion-eliciting stimulus to alter one's emotional response to that stimulus (Gross, 2013). This strategy can take different forms, with *reinterpretation* and *distancing* representing two commonly studied forms of reappraisal (Ochsner & Gross, 2008). Reinterpretation involves reconstructing the cause, context, or outcome of an emotional situation. For example, a person presented with a picture of a family losing their house in flood could think—rather than how devastating the event is—that it is lucky no one was injured and that the family may feel closer after this experience. Alternatively, distancing involves adopting an objective and detached perspective to an emotional situation. Using the same example, a person could reduce their self-proximity to the flood event by imagining it happened a long time ago or focusing on how it is personally irrelevant. Regardless of the method of implementation, reappraisal is effective in decreasing the subjective

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experience and behavioral expression of emotion (Gross, 2002; Webb et al., 2012).

At the other end of the regulation spectrum, suppression involves the effortful inhibition of emotion expression (Gross, 2013). For example, a person who loses a competition may try to hide their sadness and disappointment in order to appear calm and unbothered. Suppression is thought to be a less adaptive strategy than reappraisal in that it exacerbates physiological arousal but does not always reliably decrease negative affect in the moment (Gross & Levenson, 1997; Kalokerinos et al., 2015; cf. Jones et al., 2024).

Emotion Regulation and Memory

In addition to the affective consequences of reappraisal and suppression, research has investigated their effects on cognition and on memory specifically. Early theorizing on memory effects of emotion regulation came from the original process model (Gross, 1998). As a response-focused strategy, suppression requires ongoing self-monitoring and self-correcting effort to narrow the discrepancy between current and desired behavioral expressions. This greater self-regulatory effort is argued to reduce the cognitive resources available for memory encoding (Carver & Scheier, 1981; Ellis & Ashbrook, 1989), thus impairing memory. As an antecedent-focused strategy, reappraisal is thought to reframe an emotion-eliciting situation before emotional responses are fully activated, thus rendering the need for ongoing self-regulation unnecessary (Richards, 2004), freeing up memory capacity. This theorizing gave rise to the hypothesis that suppression should impair memory while reappraisal should leave memory intact (Richards & Gross, 2000).

Suppression and Memory

Some findings show that suppression impairs memory relative to a no-regulation control condition. In a seminal study examining the relationship between emotion regulation and memory, Richards and Gross (2000) asked participants to view a film clip that elicited negative emotion, to which they either suppressed or did not regulate their emotions. Results showed that participants who were instructed to suppress their emotions remembered fewer auditory and visual details about the film than those who did not regulate (Richards & Gross, 2000). Other researchers found a visual memory impairment effect of suppression persisted up to 1 week after emotion induction (Dunn et al., 2009). In a lab setting, Egloff et al. (2006) found that spontaneous implementation of suppression when giving a speech was associated with impaired memory for speech content. In a more naturalistic setting, Richards et al. (2003) found romantic couples who suppressed their emotions when discussing relationship conflicts showed poorer recall of conversational utterances compared with those who reappraised.

Other findings have been less consistent regarding the impact of suppression on memory. For instance, Hayes et al. (2010) found that suppression did not affect memory for images relative to a "view naturally" control condition (using images normed to American samples; see also R. C. Knight & Emery, 2022). Similarly, Wang et al. (2017) found no significant difference between suppression and control on memory for images normed to Chinese samples. Lin et al. (2022) exposed participants to a social stress test in which they gave a speech and performed challenging arithmetic in front of a disapproving panel. The authors found no difference in memory of details in the testing room between participants instructed to use suppression versus an uninstructed control condition during the stress test. Thus, while some studies suggest suppression has the capacity to impair memory, others do not.

Reappraisal and Memory

In the case of reappraisal, the literature is divided on whether this strategy improves memory or does not affect it relative to control. For example, some researchers have found that reappraisal has no effect on memory for emotion-eliciting stimuli (Kim & Hamann, 2012; M. Knight & Ponzio, 2013; Leventon & Bauer, 2016; Sheppes & Meiran, 2007), concurrent auditory information (Richards & Gross, 2000), or spontaneous speech materials (Egloff et al., 2006). Yet, other work has shown that reappraisal *enhances* memory for emotional stimuli (Davis & Levine, 2013; Dillon et al., 2007; Richards & Gross, 2000; Wang et al., 2017); an effect that remains in delayed recognition tasks (Hayes et al., 2010; Kim & Hamann, 2012).

These contradictory empirical findings mirror conflicting theories on the nature of the cognitive demands of reappraisal. Early theorizing suggested that because it changes emotions before they are fully formed, reappraisal requires minimal ongoing cognitive effort and therefore may not affect memory (Richards, 2004). An alternative perspective holds that reappraisal is indeed cognitively costly, but may facilitate or have no impact on memory (Sheppes & Meiran, 2008; Urry et al., 2009). This account is consistent with the level-of-processing framework (Craik & Lockhart, 1972), suggesting the deep elaborative processing that occurs during reappraisal lays down deeper memory traces for emotional stimuli (Dillon et al., 2007).

Taking a Closer Look

Theorizing and some empirical findings have coalesced around an idea in the literature that suppression impairs memory for emotional stimuli while reappraisal does not. And yet, the population of studies that examined the impact of emotion regulation on memory has not always yielded consistent results. Some studies have found suppression impairs memory relative to control (e.g., Bonanno et al., 2004; Richards & Gross, 2000) while others have not (e.g., Hayes et al., 2010; Wang et al., 2017). Likewise, evidence exists to suggest that reappraisal enhances (e.g., Dillon et al., 2007), has no effect on (e.g., Kim & Hamann, 2012) or impairs memory (e.g., Ortner & de Koning, 2013). For a summary of findings in the existing literature, see Table 1.

A closer examination suggests that these discrepancies may have three causes. First, studies vary in how they have approached memory, and it is possible that emotion regulation affects verbal and nonverbal memory differently. For instance, some studies show an effect of suppression on verbal memory (e.g., Egloff et al., 2006; Richards & Gross, 2000; though cf. Ortner & de Koning, 2013), while others show a null effect on nonverbal memory (e.g., Hayes et al., 2010; R. C. Knight & Emery, 2022; Wang et al., 2017; though cf. Dunn et al., 2009). Theorizing drawn from self-regulation theory (Carver & Scheier, 1981) suggests that the self-monitoring involved in suppression may incorporate elements of subvocalization (e.g., "Am I showing how I feel? I must remember not to show emotion"). This internal dialog may interfere with the encoding of verbal

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Table 1
Summary of Studies Examining the Impact of Suppression and Reappraisal on Memory

Author	Stimulus	Valence	Memory type	Finding
Bonanno et al. (2004)	Images	Neg vs. Pos	Nonverbal	Suppression < **control**
Davis and Levine (2013)	Film clips	Neg	Verbal + nonverbal	**Reappraisal (X)** = **reappraisal (↓)** > control
Dillon et al. (2007)$_a$	Images	Neg vs. Neu	Nonverbal	**Reappraisal (↑)** = **reappraisal (↓)** > control
Dillon et al. (2007)$_b$	Images	Neg vs. Neu	Nonverbal	**Reappraisal (↑)** > suppression = control
Dunn et al. (2009)	Film clips	Neg	Verbal	Suppression < **control**
Hayes et al. (2010)	Images	Neg	Nonverbal	**Reappraisal (X)** > suppression = control
Katsumi et al. (2020)	Images	Neg vs. Neu	Nonverbal	Suppression < **control** (younger adults only)
Katsumi and Dolcos (2020)	Images	Neg vs. Neu	Nonverbal	Suppression < **control** (negative images only)
Kim and Hamann (2012)	Images	Neg vs. Neu	Nonverbal	Reappraisal (↑) = reappraisal (↓) = control
R. C. Knight and Emery (2022)	Images	Neg vs. Pos	Nonverbal	Suppression = control
M. Knight and Ponzio (2013)$_a$	Images	Neg vs. Pos vs. Neu	Nonverbal	**Reappraisal (↑)** > control > reappraisal (↓)
M. Knight and Ponzio (2013)$_b$	Images	Neg vs. Pos vs. Neu	Nonverbal	**Reappraisal (↑)** > control > reappraisal (↓)
Leventon et al. (2018)$_a$	Images	Neg vs. Pos vs. Neu	Nonverbal	Reappraisal (↓) = reappraisal (→)
Leventon et al. (2018)$_b$	Images	Neg vs. Pos vs. Neu	Nonverbal	**Reappraisal (↓)** = **reappraisal (→)** > control
Leventon and Bauer (2016)	Images	Neg vs. Pos vs. Neu	Nonverbal	Reappraisal (X) = control
Lin et al. (2022)	Stress test	Neg	Nonverbal (recall)	**Reappraisal (X)** > control = suppression ("central" *items only*)
	Stress test	Neg	Nonverbal (recog)	Reappraisal = control = suppression
Martinez et al. (2023)	Images	Neg vs. Neu	Nonverbal	Reappraisal (↑) = reappraisal (X) = control
Moore and Zoellner (2012)	Film clips	Neg	Verbal + nonverbal	Suppression = control
Ortner and de Koning (2013)	Images	Pos vs. Neu	Verbal	Reappraisal (↑) = suppression < **control**
	Images	Pos vs. Neu	Nonverbal	**Reappraisal (↑)** > suppression = control (when positive images were presented first only)
Parsafar and Davis (2019)a	Film clips	Neg	Verbal	Reappraisal = control
Parsafar and Davis (2019)b	Film clips	Neg	Verbal	**Reappraisal (↓)** > reappraisal (X) = control
Richards and Gross (1999)	Images	Neg	Verbal + nonverbal	Suppression < **control**
Richards and Gross (2000)$_a$	Film clips	Neg	Verbal + nonverbal	Suppression < **control**
Richards and Gross (2000)$_b$	Images	Neg	Verbal	Suppression < **control** = **reappraisal (X)**
	Images	Neg	Nonverbal	**Reappraisal (X)** > control = suppression (high emotion images only)
Richards et al. (2003)	Conflict	Neg	Verbal	**Reappraisal (↓)** > suppression = control
Sheppes and Meiran (2007)$_a$	Film clips	Neg	Verbal	Reappraisal (X) = control
Wang et al. (2017)	Images*	Neg vs. Pos vs. Neu	Nonverbal	**Reappraisal (↑)** = **reappraisal (X)** > suppression = control
Yan et al. (2022)	Images*	Neg	Nonverbal	Suppression < **control** = reappraisal (↓ or X)
Yeh et al. (2020)	Images	Neg	Nonverbal	**Reappraisal (↓)** > control

Note. Subscripts (a, b) indicate multiple studies in one article. For significant effects, the best performing condition (i.e., condition showing the best memory) is bolded. In the stimulus column, all image stimuli drawn from International Affective Picture System except two studies designated * which used images from Chinese Affective Picture System. Reappraisal↑ = reinterpretation reappraisal tactic to intensify emotion; Reappraisal↓ = reinterpretation reappraisal tactic to diminish emotion; Reappraisal (→) reappraisal instruction to maintain emotion; Reappraisal (X) = distancing reappraisal tactic to remain psychologically removed. Neg = negative stimuli; Neu = neutral stimuli; Pos = positive stimuli. Recall = recall memory; recog = recognition memory.

information in particular (Richards & Gross, 2000). An alternative perspective is that suppression directs attention toward internal states, thus diverting attention away from the external environment (Ellis & Ashbrook, 1989). As such, suppression would be expected to impair memory for *all* stimuli, verbal and nonverbal alike. However, existing studies do not always directly compare verbal and nonverbal memory in the same design, making it difficult to reach a conclusion about the impact of memory type.

Second, studies vary in the valence of the emotion that is regulated, and it is possible that regulation of positive and negative emotion affect memory differently. Indeed, research shows positive and negative mood affect memory capacity differently (Osaka et al., 2013), with positive mood enhancing verbal and spatial working memory (Storbeck & Maswood, 2016). The possibility that valence plays a role is also borne out in existing findings of the impact of emotion regulation on memory. For instance, Richards and Gross (2000) examined the effect of regulating *negative* emotion on verbal and nonverbal memory, concluding that suppression impaired verbal memory (Studies 1 and 2) and reappraisal enhanced nonverbal memory (though only for highly emotional stimuli; Study 2). However, another study using a similar paradigm focusing on *positive* emotion found that suppression had no effect on memory, while reappraisal was associated with worse verbal memory and better nonverbal memory (Ortner & de Koning, 2013). These discrepancies warrant a systematic investigation of whether the valence of the emotion being regulated "matters" for its effect on memory.

Third and finally, existing studies investigating emotion regulation and memory may be underpowered, making it difficult to interpret mixed results (Button et al., 2013). For instance, one study in the original investigation of emotion regulation and memory for negative stimuli (Richards & Gross, 2000) tested 83 people in a three-level between-subjects design; a sensitivity analysis shows the study was powered to detect a medium–large effect of strategy on

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memory ($f = .34$). A more recent extension using positive stimuli (Ortner & de Koning, 2013) tested 65 people in a three-level between-subjects design and was powered to detect a large effect of strategy on memory ($f = .39$). By way of comparison, the effect of restricting sleep (3 hr vs. 7 hr) has a small effect on memory (Crowley et al., 2024). Indeed, large effect sizes in general are very uncommon in psychology (Funder & Ozer, 2019). In sum, well-powered tests are needed that capture the regulation of positive and negative emotion applied to verbal and nonverbal memory.

The Present Research

To resolve inconsistencies in the literature, we conducted four highly powered experiments (*N*s > 1,000) testing whether reappraisal and suppression impact on verbal and nonverbal memory relative to control. In addition to testing the impact of these emotion regulation strategies on verbal and nonverbal memory, we used an experimental protocol that allowed us to probe the nature of the cognitive resources impacted by emotion regulation. This protocol is known as a dual-task memory paradigm.

A dual-task memory paradigm involves performing a goal-relevant primary task alongside a concurrent task (Fisk et al., 1986). This method draws on theories of resource allocation (Wickens, 1991), which suggest that different cognitive tasks compete for limited mental resources when performed simultaneously. The result of this competition is that an effortful task may tax cognitive resources that could otherwise be allocated for concurrent task processing (Ellis & Ashbrook, 1989; Wickens, 1991). In the context of emotion regulation, a dual-task design may involve asking participants to regulate emotions induced by emotional images, while listening to a series of words at the same time. In the dual-task paradigm, the cognitive resources required for the goal-relevant task (e.g., the act of regulating emotions evoked by the images) is indicated by performance on the concurrent task (e.g., hearing words). This approach therefore provides insight into the cognitive resources required to enact reappraisal and suppression.

In our experimental protocol, participants were instructed to reappraise, suppress, or not regulate emotions induced by emotional images. At the same time, participants listened to words played over headphones. Subsequently, we administered a surprise recognition memory test for images and words. We conducted two experiments using negative stimuli (Experiments 1 and 3) and two experiments using positive stimuli (Experiments 2 and 4). To examine different forms of reappraisal, in Experiments 1 and 2, we operationalized this strategy as reinterpretation and in Experiments 3 and 4 as distancing. All studies were preregistered and powered over 80% to detect small to very small effects.

Based on past theorizing (Richards, 2004), we hypothesized nonverbal memory for images would be worse in the suppression condition relative to control and reappraisal conditions (H1). Based on past findings (Ortner & de Koning, 2013; Richards & Gross, 2000), we hypothesized verbal memory for words would be worse in both the reappraisal and suppression conditions relative to control (H2). We did not have separate hypotheses for positive and negative emotion, given limited studies using positive stimuli.

In Experiments 1 and 2, we also manipulated whether participants heard neutral or valenced words while viewing the images. We did this to test whether the valence of verbal information moderated the relationship between emotion regulation and memory. Our original exploratory reasoning was that words that were conceptually related to the images (i.e., of the same valence) may influence the elaborative encoding that occurs during reinterpretation-based reappraisal (as per Ortner & de Koning, 2013). That is, if reinterpretation encourages deep encoding as people engage with the images, words that are conceptually related to the content of the images (i.e., valenced words) may facilitate the encoding process, thus improving memory. This was a tentative possibility, and as such we made no a priori predictions about this additional manipulation. Further, as it had no effect, we omitted the word valence manipulation in Experiments 3 and 4.

Transparency and Openness

All experiments were preregistered. We report how we determined our sample size, all data exclusions, all manipulations, and all measures. Research materials are available on the Open Science Framework at https://osf.io/edtrk/, including full experimental protocols and materials, preregistrations, Supplemental Materials, and data and code for replicating analyses.

Experiment 1: Negative Emotion

Method

Participants and Design

Participants[1] were 1,055 MTurk workers (43% female, 98% English as primary language, $M_{age} = 35.54$, $SD_{age} = 10.21$, range = 18–71 years) who completed the experiment in 2019 and were paid USD $1.80. The experiment had a 2 (word type: neutral vs. negative) × 3 (emotion regulation strategy: reappraisal vs. suppression vs. control) between-subjects design. The dependent variables were recognition memory for images and words. A sensitivity analysis revealed the experiment was powered at 80% to detect a small interaction effect ($f = .10$). The experiment was preregistered, and all study materials, data, and code are available at https://osf.io/edtrk/.

Procedure

Participants provided informed consent and rated their baseline emotion experience. After being randomly assigned to experimental condition and reading about their designated strategy, participants completed a practice trial to gain experience with the task and the regulation strategy. Participants in the reappraisal condition were asked to down-regulate their response to the picture using a reinterpretation tactic. Participants in the suppression condition were asked to control their facial expressions to the picture. Participants in the control condition received no emotion regulation instructions.

In the practice trial, participants were asked to implement the strategy while viewing an image for 10 s, with a word played 3 s after image onset. Participants were asked to confirm that they had seen an image and heard a word, and completed a manipulation

[1] Based on preregistered criteria, we excluded 134 participants (11% of the original sample, which was $N = 1,189$) who experienced technical difficulties, failed attention checks, failed all filler questions, had hearing difficulties, and were color blind. Effects using the full data set with these participants included are reported in Supplemental Materials and show substantively identical results (Supplemental Table S1).

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check asking how they implemented the strategy. If participants implemented the strategy successfully according to the check, they progressed in the study, and if not were provided corrective feedback before progressing.

Participants viewed 20 images and listened to 20 words while regulating their emotions as instructed. Image stimuli featured all negative pictures. Each image was presented for 10 s, and the word was pronounced 3 s after image onset. Order of images was randomized for each participant. Twenty target words were randomly extracted from a list of 40 words and the remaining 20 words were used as lures in the recognition task. Participants in the negative word condition heard only negative words; participants in the neutral word condition heard only neutral words. The negative words were conceptually related to the images in that they involved abstract words of the same valence as the negative images. The neutral words were unrelated to the images.

Following encoding, participants rated their postmanipulation emotion experience and completed manipulation checks, attention checks, and filler questions. Then, participants completed a surprise recognition memory test for images and words. Participants were asked to identify a target image from four alternatives arranged in random positions displayed in a 2 × 2 grid. Participants were then presented with 40 words and asked to indicate whether they heard the word during the trial phase. All image and word stimuli were counterbalanced.

Materials and Measures

Word Stimuli. We selected a total of 80 words (40 negative, 40 neutral) from the Affective Norms for English Words (Bradley & Lang, 1999). Each word had normed ratings for arousal (1 = *lowest arousal*, 9 = *highest arousal*) and valence (1 = *most negative*, 9 = *most positive*). Negative words were selected based on their low valence ratings ($M_{\text{negative}} = 2.28$, $SD = 0.45$) compared with the neutral words ($M_{\text{neutral}} = 5.52$, $SD = 0.29$). Negative and neutral words did not differ on word frequency, $t(78) = 0.12$, $p = .902$, nor word length, $t(78) = -0.07$, $p = .942$, but differed on arousal rating, $t(78) = 6.45$, $p < .001$.

Image Stimuli. We chose 20 negative images from the International Affective Picture System (Lang et al., 2005). The mean valence (1 = *most negative* to 9 = *most positive*) and arousal ratings (1 = *lowest arousal* to 9 = *highest arousal*) were 3.44 ($SD = 0.27$) and 5.21 ($SD = 0.58$), respectively. We generated three lures from each original image, including being (a) flipped horizontally, (b) color adjusted, or (c) both flipped horizontally and color adjusted (see Supplemental Materials, e.g., all lures available on the Open Science Framework at https://osf.io/edtrk/).

Emotion Regulation Instructions. We adapted the correctional feedback approach to training the emotion regulation strategies from Sheppes et al. (2011). Verbatim manipulation instructions are included in Supplemental Materials. Participants in the reappraisal condition were asked to change the meaning of what was happening in the picture to make themselves feel less negative about it. Participants in the suppression condition were asked to control their facial expressions in such a way that others would not be able to tell they were feeling anything at all. Control participants received no emotion regulation instructions.

Recognition Memory. For image and word recognition memory, correct responses were scored as 1 and incorrect responses as 0.

Emotion Experience. Participants rated their emotion experience using 18 descriptors adapted from Gross and Levenson (1997; 1 = *not at all*, 7 = *very much*). Effects on emotion experience for all experiments are reported in Supplemental Tables S3 and S4.

Manipulation Checks. Three items adapted from Gross (1998) examined successful reappraisal ("When viewing the images, I tried to think about what was happening in the picture in a new way so I felt less negative"), suppression ("When viewing the images, I tried to keep my face neutral so no one would know how I was feeling"), or control instruction implementation ("I viewed the images as I normally would, without using any particular strategy"; 1 = *strongly disagree*, 7 = *strongly agree*).

Attention Checks and Filler Questions. Attention checks were two multiple-choice questions on general knowledge to ensure data quality (Maniaci & Rogge, 2014). Filler questions were three moderate-difficulty multiple-choice math questions to avoid unconscious rehearsal effects before the memory test (Bower & Gilligan, 1979).

Results

Manipulation Checks

Table 2 shows descriptive and inferential statistics for the manipulation check and memory variables. Separate 2 × 3 analyses of variance revealed the expected main effects of emotion regulation on the separate emotion regulation checks (Fs > 221.11, ps < .001).

Table 2
Effect of Emotion Regulation Manipulation on Manipulation Checks and Memory for Experiments 1 and 2

	Experiment 1 (negative emotion)					Experiment 2 (positive emotion)				
Variable	Suppression	Reappraisal	Control	$F(df)$	η_p^2	Suppression	Reappraisal	Control	$F(df)$	η_p^2
Manipulation checks				(2, 1049)					(2, 1027)	
Suppression check	**6.23_a (1.24)**	3.43_b (2.08)	3.11_c (2.01)	303.01^{***}	.31	**6.37_a (1.16)**	3.35_b (2.12)	2.95_c (2.07)	370.91^{***}	.42
Reappraisal check	3.17_b (2.03)	**6.16_a (1.15)**	3.25_b (1.92)	334.75^{***}	.32	2.37_b (1.92)	**6.20_a (1.28)**	2.37_b (1.92)	550.30^{***}	.52
Control check	3.09_b (2.11)	2.39_c (1.86)	**5.30_a (1.86)**	221.11^{***}	.24	2.89_b (2.02)	2.64_b (2.08)	**6.06_a (1.30)**	366.52^{***}	.42
Memory				(2, 1051)					(2, 1029)	
Nonverbal	0.49_a (0.19)	0.50_a (0.19)	0.51_a (0.19)	1.10	<.01	0.52_b (0.22)	**0.56_a (0.23)**	$0.54_{a,b}$ (0.23)	4.63^{*}	.01
Verbal	0.69_b (0.01)	0.70_b (0.01)	**0.74_a (0.01)**	6.72^{**}	.01	0.73_a (0.01)	0.72_a (0.01)	0.74_a (0.01)	1.23	<.01

Note. Different subscripts (a, b) within each row indicate means across experimental conditions are significantly different at $p < .05$. Values in parentheses indicate standard deviation for manipulation check results and standard error for memory results. The highest rating for all significant effects is bolded for ease of interpretation.
$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

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Pairwise comparisons showed the manipulations were successful: Participants in the suppression condition reported using suppression more than participants in the reappraisal and control conditions, and participants in the reappraisal condition reported using reappraisal more than participants in the suppression and control conditions. There were no significant main effects of word type and no interaction (Fs < 2.15, ps > .118).

Memory

After initial analysis as per our original preregistration, we updated our analytic strategy to use multilevel modeling to account for the nested nature of the data. We preregistered this updated analysis plan prior to performing multilevel analyses, available at https://osf.io/edtrk/. We analyzed recognition memory for images and words using mixed effects models with *lme4* (Bates et al., 2015) and *afex* (Singmann et al., 2018) packages in R. Pairwise comparison tests were conducted using the R package *emmeans* (Lenth et al., 2018).

All models included fixed effects of experimental factors (strategy and word type). We privilege the main effects of emotion regulation strategy when discussing results, but footnote rare cases where we observed a significant main effect of word type. We conducted exploratory analyses of the interaction between emotion regulation strategy and word type. However, none of these effects were significant (see Supplemental Table S5).

We added image arousal, image valence, word arousal, and word valence ratings to the model factorially as fixed effects and retained these variables if they significantly improved model fit using chi-squared significance tests. We initially preregistered including participant, image stimuli, and word stimuli as random effects for all models. However, we removed random effects that did not correspond to the relevant memory type in the face of convergence problems (Barr et al., 2013). Thus, nonverbal memory models included participant and image stimuli as random effects, and verbal memory models included participant and word stimuli as random effects. Final model specifications for all tests are outlined in Supplemental Table S5.

Nonverbal Memory. Word valence improved model fit relative to a model with only experimental factors, $\chi^2(1) = 5.24$, $p = .022$. Thus, the final model included fixed effects of experimental factors and word valence ratings, as well as random intercepts for participants and image stimuli. Contrary to H1, there was no significant main effect of emotion regulation strategy, $F(2, 1051) = 1.10$, $p = .334$, $\eta_p^2 < .01$.[2]

Verbal Memory. No additional fixed effect improved model fit (ps > .070). Thus, our final model included only fixed effects of experimental factors and random intercepts for participants and word stimuli. There was a significant main effect of emotion regulation strategy, $F(2, 1051) = 6.72$, $p = .001$, $\eta_p^2 = .01$. Supporting H2, pairwise comparisons revealed participants in the control condition recognized words more accurately than participants in the suppression, $p = .003$, $d = 0.11$, and reappraisal conditions, $p = .008$, $d = 0.09$.

Experiment 2: Positive Emotion

Method

Participants and Design

Participants[3] were 1,033 MTurk workers (41% female, 99% English as primary language, $M_{age} = 36.13$, $SD_{age} = 10.59$, range = 19–69 years) who did the experiment with the same specifications as Experiment 1 but featuring positive instead of negative stimuli. The experiment had a 2 (word type: neutral vs. positive) × 3 (emotion regulation strategy: reappraisal vs. suppression vs. control) between-subjects design. The dependent variables were recognition memory for images and words. A sensitivity analysis revealed the experiment was powered at 80% to detect a small interaction effect ($f = .10$). The experiment was preregistered, and all study materials, data, and code are available at https://osf.io/edtrk/.

Procedure

The procedure was identical to Experiment 1.

Materials and Measures

Materials were identical to Experiment 1, except for the following changes. First, while the suppression instructions were the same as in Experiment 1, we adapted the wording of the reappraisal instructions from Ortner and de Koning (2013), who studied the regulation of positive emotion. Verbatim instructions are included in Supplemental Materials. Participants in the reappraisal condition were asked to change the meaning of what was happening in the picture so as to make themselves feel *more negative* about it. Reflecting this change, the reappraisal manipulation check to assess for correct implementation of this strategy was changed to: "When viewing the images, I tried to think about what was happening in the picture in a new way so I felt more negative," scored on a scale from 1 (*strongly disagree*) to 7 (*strongly agree*). The other manipulation checks remained the same as Experiment 1.

Second, valenced word stimuli in Experiment 2 consisted of 40 positive words, with a mean valence rating of 7.60 ($SD = 0.44$; 1 = *most negative*, 9 = *most positive*). Positive and neutral words did not differ on CELEX word frequency, $t(78) = -1.44$, $p = .154$, nor word length, $t(78) < 0.01$, $p > .999$, but differed on arousal rating, $t(78) = -5.82$, $p < .001$.

Results

Manipulation Checks

Table 2 shows descriptive and inferential statistics for the manipulation check and memory variables. The regulation manipulations were successful (Fs > 366.52, ps < .001).[4]

Memory

The analytic strategy was identical to Experiment 1 (see Supplemental Table S5 for final model specifications).

[2] There was a main effect of word type ($p = .022$): participants who heard negative words showed better memory for images than participants who heard neutral words.

[3] Based on preregistered criteria, we excluded 169 participants (14% of the original sample, which was $N = 1,202$) who experienced technical difficulties, failed attention checks, failed all filler questions, had hearing difficulties, and were color blind. Effects using the full data set with these participants included are reported in Supplemental Materials and show substantively identical results (Supplemental Table S1).

[4] There was a significant main effect of word type on the control check ($p = .019$) but no other significant main effects and no interactions (Fs < 1.97, ps > .140).

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Nonverbal Memory. No additional fixed effect significantly improved model fit ($ps > .156$), thus our final model included only fixed effects of experimental factors and random intercepts for participants and image stimuli. Partially supporting H1, there was a significant main effect of emotion regulation strategy, $F(2, 1029) = 4.63$, $p = .010$, $\eta_p^2 = .01$, such that participants in the suppression condition recognized images less accurately than participants in the reappraisal condition, $p = .007$, $d = 0.08$, although did not differ from participants in the control condition, $p = .163$, $d = 0.07$.

Verbal Memory. Image valence significantly improved model fit relative to the model with only experimental factors, $\chi^2(1) = 5.82$, $p = .016$. Thus, the final model included fixed effects of experimental factors and image valence ratings, as well as random intercepts for participants and word stimuli. Contrary to H2, there was no significant main effect of emotion regulation strategy, $F(2, 1029) = 1.23$, $p = .188$, $\eta_p^2 < .01$.

Discussion

Experiments 1 and 2 tested whether regulating negative and positive emotion, respectively, influenced nonverbal memory for images and verbal memory for words. Using a dual-task memory paradigm, we found effects on both types of memory as a function of regulation strategy. In Experiment 1, suppression *and* reappraisal of negative emotion impaired memory for words relative to control. In Experiment 2, reappraisal of positive emotion improved memory for images relative to suppression but not control. Across both experiments, we found no evidence that the valence of the word presented alongside images influenced memory effects.

These interim findings suggest the impact of suppression and reappraisal on memory may depend on both emotion valence and memory type. Namely, suppression impaired verbal memory for negative words (relative to control) and impaired nonverbal memory for positive images (relative to reappraisal). Put otherwise, the effect of reappraisal was to improve nonverbal memory for positive images relative to suppression. This latter effect is in keeping with a finding by Ortner and de Koning (2013) as well as theorizing that reinterpretation-based reappraisal involves deep elaborative processing that should improve memory for emotional stimuli (Dillon et al., 2007). Probing for boundary conditions to the reappraisal effect, we tested a different form of reappraisal in Experiments 3 and 4. These studies focused on a distancing reappraisal tactic, as well as aiming to replicate the effects of suppression.

Experiment 3: Negative Emotion

Method

Participants and Design

Participants[5] were 1,106 Prolific workers (55% female, 99% English as primary language, $M_{age} = 40.78$, $SD_{age} = 12.78$, range = 18–83 years) who completed the experiment in 2021 and were paid £1.50. The study used negative stimuli and had a three-level (emotion regulation strategy: reappraisal vs. suppression vs. control) between-subjects design. The dependent variables were recognition memory for images and words. A sensitivity analysis revealed the experiment was powered at 80% to detect a very small one-way, three-level omnibus effect of emotion regulation strategy ($f = .08$). The experiment was preregistered, and all study materials, data, and code are available at https://osf.io/edtrk/.

Procedure

The procedure was largely identical to Experiments 1 and 2, but did not manipulate word type (i.e., all words were negative) and focused instead on the main effect of emotion regulation. In addition, we preregistered further steps to ensure data quality. Before beginning the experimental protocol, participants were presented with a test image and a test word and asked to describe those stimuli in a few words. We reviewed the responses, and excluded from analyses participants who did not correctly identify the test image or word. In addition, we monitored remotely whether participants navigated away from the browser containing the experiment. If the participant tabbed to another browser for more than 5 s, the screen displayed a pop-up message: "It seems like you tabbed away from the experiment. Please stay focused during the task." We additionally excluded from analysis participants who tabbed away from the experiment browser on more than 50% of trials.

Materials and Measures

Materials and measures were largely identical to Experiment 1 except for the following changes. First, we did not manipulate word type, presenting only negative words to participants. Second, we added an extra attention check question asking participants what they had for dinner yesterday. Third, we changed the nature of the reappraisal manipulation. Experiment 1 asked participants to adopt a reinterpretation reappraisal tactic in which they altered the meaning of what was happening in the image to feel less negative. In Experiment 3, we asked participants to adopt a distancing reappraisal tactic so they remained distant from the image in such a way that they did not feel any emotion. In doing so, we adhered closely to the instruction for the reappraisal condition from the original Richards and Gross's (2000) study. Reflecting this change, we altered the wording of the reappraisal manipulation check to read: "When viewing the images, I tried to think about what was happening in the picture in an objective way so I felt nothing at all," scored on a scale from 1 (*strongly disagree*) to 7 (*strongly agree*). The other manipulation checks remained the same as Experiments 1 and 2.

Results

Manipulation Checks

Table 3 shows descriptive and inferential statistics for manipulation checks and memory. The emotion regulation manipulations were successful ($Fs > 168.50$, $ps < .001$).

[5] Based on preregistered criteria, we excluded 85 participants (7% of the original sample, which was $N = 1{,}191$) who experienced technical difficulties, failed attention checks, failed all filler questions, had hearing difficulties, were color blind, and tabbed away from the experiment on more than 50% of the trials. Effects using the full data set with these participants included are reported in Supplemental Materials and show substantively identical results (Supplemental Table S2).

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Table 3
Effect of Emotion Regulation Manipulation on Manipulation Checks and Memory for Experiments 3 and 4 (Standard Deviation in Parentheses)

Variable	Experiment 3 (negative emotion)					Experiment 4 (positive emotion)				
	Suppression	Reappraisal	Control	$F(df)$	η_p^2	Suppression	Reappraisal	Control	$F(df)$	η_p^2
Manipulation checks				(2, 1103)					(2, 1097)	
Suppression check	**6.37_a (1.02)**	4.59_b (1.97)	3.10_c (1.78)	374.80***	.40	**6.42_a (0.99)**	4.66_b (1.89)	2.72_c (1.79)	482.50***	.47
Reappraisal check	4.39_b (1.94)	**5.86_a (1.27)**	3.68_c (1.63)	168.50***	.23	4.48_b (1.85)	**6.17_a (1.16)**	3.33_c (1.78)	282.50***	.34
Control check	2.62_c (1.71)	3.38_b (1.91)	**5.53_a (1.49)**	288.50***	.34	2.71_c (1.77)	3.06_b (1.98)	**5.84_a (1.35)**	356.80***	.39
Memory				(2, 1103)					(2, 1097)	
Nonverbal	0.55_a (0.02)	0.55_a (0.02)	0.54_a (0.02)	0.24	<.01	0.59_a (0.03)	0.61_a (0.03)	0.60_a (0.03)	1.37	<.01
Verbal	0.74_b (0.02)	$0.76_{a,b}$ (0.02)	**0.77_a (0.02)**	4.15*	.01	0.72_a (0.01)	0.74_a (0.01)	0.75_a (0.01)	1.52	<.01

Note. Different subscripts (a, b) within each row indicate means across experimental conditions are significantly different at $p < .05$. Values in parentheses indicate standard deviation for manipulation check results and standard error for memory results. The highest rating for all significant effects is bolded for ease of interpretation.
* $p < .05$. *** $p < .001$.

Memory

Our analytic strategy was preregistered, and very similar to that in Experiments 1 and 2. All models included fixed effects of the experimental factor (emotion regulation strategy). We added image arousal, image valence, word arousal, and word valence to the model factorially as fixed effects and retained these variables as fixed effects if they significantly improved model fit using chi-squared significance tests. Nonverbal memory models included participant and image stimuli as random intercepts, and verbal memory models included participant and word stimuli as random effects. Final model specifications for all tests are outlined in Supplemental Table S5.

Nonverbal Memory. No additional fixed effect (valence or arousal ratings) improved model fit (ps > .250). Our final model included a fixed effect of the experimental factor and random intercepts for participants and image stimuli. Contrary to H1, there was no significant main effect of emotion regulation strategy, $F(2, 1103) = 0.24$, $p = .785$, $\eta_p^2 < .01$.

Verbal Memory. Image arousal significantly improved model fit relative to the model with only experimental factors, $\chi^2(1) = 5.83$, $p = .016$. Thus, the final model included fixed effects of the experimental factor and image arousal ratings, as well as random intercepts for participants and word stimuli. Partially supporting H2, there was a significant main effect of emotion regulation strategy, $F(2, 1103) = 4.15$, $p = .016$, $\eta_p^2 = .01$: Participants in the control condition recognized words more accurately than participants in the suppression condition, $p = .013$, $d = 0.07$, although control participants did not differ from participants in the reappraisal condition, $p = .579$, $d = 0.02$. There was no significant difference between the reappraisal and the suppression conditions, $p = .163$, $d = 0.05$.

Experiment 4: Positive Emotion

Method

Participants and Design

Participants[6] were 1,100 Prolific workers (59% female, 99% English as primary language, $M_{age} = 39.25$, $SD_{age} = 13.01$, range = 18–78 years) who did the experiment with the same specifications as Experiment 3, but featuring positive stimuli. The experiment had a three-level (emotion regulation strategy: reappraisal vs. suppression vs. control) between-subjects design. The dependent variables were recognition memory for images and words. A sensitivity analysis revealed the experiment was powered at 80% to detect a very small one-way three-level omnibus effect of emotion regulation strategy ($f = .08$). The experiment was preregistered, and all study materials, data, and code are available at https://osf.io/edtrk/.

Procedure

The procedure was identical to Experiment 3.

Materials and Measures

Materials were very similar to Experiment 2 except for the following changes. We did not manipulate word type, presenting only positive words to participants, and added an extra attention check question asking participants what they had for dinner yesterday. As in Experiment 3, we changed the nature of the reappraisal manipulation. Experiment 2 asked participants to adopt a reinterpretation reappraisal tactic in which they altered the meaning of what was happening in the (positive) image so as to feel *more negative* about it (adapted from Ortner & de Koning, 2013). In contrast, in Experiment 4, we asked participants to adopt a distancing reappraisal tactic, so that they remained distant from what was happening in the (positive) image in a way that they did not feel any emotion about it (identical wording to Experiment 3 and adapted from Richards & Gross, 2000).

[6] Based on preregistered criteria, we excluded 85 participants (7% of the original sample, which was $N = 1{,}185$) who experienced technical difficulties, failed attention checks, failed all filler questions, had hearing difficulties, were color blind, and tabbed away from the experiment for more than 50% of the trials. Coincidentally, this is the same number of participants excluded in Experiment 3. Effects using the full data set with these participants included are reported in Supplemental Materials and show substantively identical results (Supplemental Table S2).

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Results

Manipulation Checks

Table 3 shows descriptive and inferential statistics for the manipulation check and memory variables. The emotion regulation manipulations were successful (Fs > 282.50, ps < .001).

Memory

The analytic strategy was preregistered and identical to Experiment 3.

Nonverbal Memory. No additional fixed effect significantly improved model fit (ps > .213). Contrary to H1, there was no significant main effect of emotion regulation strategy, $F(2, 1097) = 1.37$, $p = .256$, $\eta_p^2 < .01$.

Verbal Memory. No additional fixed effect significantly improved model fit (ps > .231). Contrary to H2, there was no significant main effect of emotion regulation strategy, $F(2, 1097) = 1.52$ $p = .219$, $\eta_p^2 < .01$.

Discussion

Experiments 3 and 4 tested whether suppression or reappraisal influenced nonverbal memory for images and verbal memory for words. We again found different effects as a function of strategy. In Experiment 3, suppression of negative emotion impaired memory for words relative to control, but not reappraisal. In Experiment 4, we found no effects of either reappraisal or suppression of positive emotion on either type of memory. Considering the results together, there is little evidence that reappraising using a distancing tactic influences memory, and some evidence that suppression of negative emotion may worsen verbal memory relative to control.

General Discussion

This research tested the assumption that suppression—not reappraisal—impairs memory. In doing so, we aimed to resolve three inconsistencies in prior work. First, mixed findings in existing research may have stemmed from the fact that emotion regulation affects *either* verbal *or* nonverbal memory or that it affects both forms of memory in different directions. Second, existing research has tended to focus on negative emotion, rather than assessing positive and negative emotion regulation systematically. Third, and compounding these other issues, existing research has tended to be underpowered. Addressing these issues, we conducted four well-powered experiments to test the effect of regulation of negative emotion (Experiments 1 and 3) and positive emotion (Experiments 2 and 4) on nonverbal and verbal memory. Although most findings were null, we did observe a consistent effect across the studies, such that suppression of negative emotion reduced verbal memory relative to control. Study findings are summarized in Table 4.

Inconsistent Effects of Emotion Regulation on Nonverbal Memory

We found no consistent effects of emotion regulation on nonverbal memory for images. Memory for images was not affected in Experiments 1 and 3 (negative emotion) or 4 (positive emotion). In an isolated effect, nonverbal memory was improved by reappraisal relative to suppression (but not control) in Experiment 2 in which positive emotion was regulated using a reinterpretation reappraisal tactic.

An Effect of Negative Suppression on Verbal Memory

Verbal memory was impaired by both emotion regulation strategies relative to control in Experiment 1 (negative emotion) and was impaired by suppression—but not reappraisal—relative to control in Experiment 3 (negative emotion). There was no effect of regulating positive emotion on verbal memory in Experiment 2 or 4. Thus, a take-away of this research is that *suppression* of *negative* emotion impairs *verbal* memory relative to control.

Theoretical and Methodological Implications

We found evidence that emotion regulation affected memory, though the effect was isolated to one emotion regulation strategy, one valence of emotion, and one form of memory. This observation is consistent with findings in a seminal article that is widely cited to support the argument that suppression worsens memory. Richards and Gross (2000) conducted two experiments investigating this question, finding that suppression of negative emotion impaired both verbal and nonverbal memory relative to control (Study 1) and impaired only verbal—not nonverbal—memory relative to control (Study 2). These findings are in line with our results, suggesting a consistent effect. However, we argue the assumption that

Table 4
Summary of Effects Across the Experiments

Comparison	Experiment 1 Negative	Experiment 2 Positive	Experiment 3 Negative	Experiment 4 Positive
Nonverbal memory				
Suppression vs. control				
Reappraisal vs. control				
Reappraisal vs. suppression		(+) $d = 0.08$		
Verbal memory				
Suppression vs. control	(–) $d = 0.11$		(–) $d = 0.07$	
Reappraisal vs. control	(–) $d = 0.09$			
Reappraisal vs. suppression				

Note. (+) improves memory; (–) impairs memory; blank = nonsignificant; d = Cohen's d effect size. Gray shaded boxes indicate an effect that replicates across two experiments.

suppression impairs memory should be updated to reflect the nuance of this effect: it is small, isolated to negative emotion, and seems to occur only for certain types of memory and relative to a no-regulation control condition.

There is a second way to interpret our results, which is that—rather than influencing verbal memory per se—suppression impairs memory for a nonfocal, secondary task (in our paradigm, hearing words) performed alongside a goal-relevant task (in our paradigm, viewing images). If suppression impairs performance on a secondary task, regardless of whether that task is verbal or nonverbal, it would suggest that the act of suppression is generally effortful, depleting cognitive resources available for a concurrent task. If suppression impairs performance on a verbal task, regardless of whether that task is focal or secondary, it would suggest that suppression taxes cognitive resources that are uniquely involved in verbal processing. Such an interpretation would be in keeping with theories of self-monitoring that propose a cognitively costly "subvocalization" dialog (Carver & Scheier, 1981; Richards & Gross, 2000). To fully probe these nuances, additional designs would be needed in which participants perform a primary verbal task alongside a secondary nonverbal task. For now, however, our results are clear that suppression of negative emotion (relative to control) impairs memory for a secondary/verbal task.

Already, the nuance in our one consistent finding raises implications for how the field approaches the question of whether emotion regulation influences memory. At their most basic, our findings argue against inclusion of a blanket statement in papers that "suppression, not reappraisal, impairs memory." Further, the finding suggests that suppressing negative emotion is different—and perhaps takes a greater toll—than suppressing positive emotion. This observation seems intuitive, and indeed research shows people report finding it more difficult to regulate negative emotion than positive emotion (Kalokerinos et al., 2015). Moreover, negative mood impairs verbal and spatial working memory relative to positive mood (Storbeck & Maswood, 2016). Hence, it seems possible that suppressing negative emotion worsens memory more than suppressing positive emotion, both because it is more challenging to do and because it taxes memory capacity to a greater degree. However, the potential for valence-based differences in emotion regulation is relatively under-theorized in the literature. The field has more work to do in articulating differences between regulating positive and negative emotion, as well as the precise mechanisms through which these types of regulation affect memory.

In contrast to suppression, reappraisal did not have consistent effects on memory relative to control. We observed two isolated effects such that reappraisal (a) worsened verbal memory relative to control (but not suppression) in Experiment 1 and (b) improved nonverbal memory relative to suppression (but not control) in Experiment 2. Neither of these findings is in keeping with original process model theorizing that reappraisal—as an antecedent-focused strategy—renders ongoing regulation unnecessary and thus should not influence memory (Richards, 2004; Richards & Gross, 2000).[7] These findings provide some support for the account that reappraisal is cognitively effortful (i.e., impairs memory for a secondary task, as shown in Experiment 1), but involves elaborative processing of emotional stimuli and thus facilitates encoding of focal stimuli (i.e., improves memory for images, as shown in Experiment 2, though the effect was only significant relative to suppression).

The two reappraisal effects we observed occurred in experiments that operationalized reappraisal using a reinterpretation tactic. That is, in Experiment 1, participants thought about negative images in a way that made them feel more positive and in Experiment 2 thought about positive images in a way that made them feel *more negative*. This latter operationalization was adopted from Ortner and de Koning (2013), who found similar results: reappraisal improved memory for positive images relative to a control condition. This corroborating evidence may suggest that the effect of positive reappraisal on nonverbal memory is robust, but unique to the reinterpretation tactic, given that positive reappraisal in the form of distancing did not show the same effect in Experiment 4. Reinterpretation as a tactic is more consistent with theoretical accounts that argue for a memory enhancing effect of reappraisal as a function of stimulus elaboration (Dillon et al., 2007). That is, by engaging deeply with the visual stimulus to change its emotional meaning, people may lay down deeper memory traces for the stimulus in question. However, this explanation does not make clear why such an effect would occur for the regulation of positive but not negative emotion and also awaits more targeted testing of the effects of different reappraisal tactics on memory.

Caveats, Limitations, and Future Directions

Constraints on Generality

Findings from the present research must be interpreted in light of its weaknesses. Notably, our design constrains us to making conclusions about the effects of certain types of reappraisal and suppression on certain types of memory. That is, we can draw strong conclusions about the effects of these strategies on recognition memory for static images and words, but these effects may not generalize to other types of memory (e.g., recall memory) or other stimuli (e.g., videos rather than images). Further, our verbal stimuli were in English, and our samples were English-speaking; it is possible our verbal memory findings may not generalize to other languages.

While we did observe some effects of emotion regulation on recognition memory, effects were small and may not be entirely robust (i.e., most p values $> .01$; Lakens, 2015). This is important to keep in mind when comparing the present findings with those of past studies, some of which were underpowered. Even assuming the obtained effect of negative suppression on verbal memory is robust, the effect size may invite people to wonder if the effect is significant but nonetheless meaningless (though under some circumstances small effects can be powerful; Prentice & Miller, 2016, and online effects are often smaller than their in-lab counterparts; Crump et al., 2013). Either way, our findings at worst call into question the generally cited assumption that suppression worsens memory while reappraisal does not, and at best argue for revising this accepted wisdom to more accurately represent the nuance in the effect. Emotion regulation researchers should strive to be precise in the predictions we make—and to test those predictions in preregistered,

[7] Note, however, that with the proposal of the extended process model (Gross, 2015), there is general theoretical agreement that reappraisal represents a range of dynamic and iterative cognitive processes, rather than a purely "antecedent-based" decision (Gross, 2015; Sheppes, 2020).

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confirmatory, well-powered studies—to gain greater clarity about the nature of those effects.

One factor that may have affected the results of our experiments is that the image stimuli were not very high in intensity and therefore may not be strong enough to yield a noticeable memory enhancement effect of reappraisal if one exists. This may be a broader issue for the field to consider, particularly for studies using image stimuli to elicit emotion. Indeed, as Levenson (2014) said: "*As long as the bulk of existing research … is plagued by problems related to unverified emotion elicitation*, *low emotional intensity … it will be impossible to reach well-founded*, *data-based conclusions*" (p. 110). It is possible that the intensity of emotion elicitation influences memory effects: a previous study found that reappraisal only enhanced memory for high but not low intensity emotional stimuli (note with caution, however, that this was a nonhypothesized finding based on a nonsignificant interaction; Richards & Gross, 2000). This possibility speaks further to the importance of explicitly stating the conditions under which memory effects are expected to emerge and testing these conditions in preregistered, confirmatory experiments.

Our experiments were systematic, following virtually the same protocol in each case. This had the benefit of allowing us to make direct comparisons between results, but limited the conclusions we can draw compared with other designs we might have adopted. For instance, a design that varied the nature of the memory task—such that participants regulated emotions induced by words while viewing unrelated images—would have afforded different inferences about the nature of the memory deficit associated with suppression (i.e., whether it is uniquely verbal in nature). Further, designs incorporating a within-subjects component would allow for a more direct comparison between strategies due to counterbalancing of emotion regulation instructions. Finally, ecologically valid designs, such as those using personally chosen film clips or experience sampling methods, would test memory effects where they really matter—in people's daily lives (e.g., Colombo et al., 2024).

The experiments were conducted online, raising concerns about whether participants were fully attentive during the experiment and whether they applied the instructed strategy effectively. We targeted this concern in five ways. First, we introduced a practice trial and provided corrective feedback to ensure participants understood the emotion regulation instructions (drawn from a similar procedure by Sheppes et al., 2011). Second, we excluded participants from analyses who failed preregistered criteria to mitigate potential issues associated with poor-quality data (Chmielewski & Kucker, 2019). Results that include these participants are reported in Supplemental Tables S1 and S2 and mirror the results reported in this article. Third, we included manipulation checks to test whether participants effectively employed the requested strategy. Fourth, given that concerns about MTurk data emerged after running Experiments 1 and 2, in Experiments 3 and 4, we recruited participants from Prolific; a site shown to yield higher quality data (Peer et al., 2022). Fifth, in Experiments 3 and 4, we introduced more stringent checks for data quality. In combination with the high power, practice trials, and successful manipulation checks in all studies, these procedures increase our confidence in the obtained results. We recommend that researchers adopt and report similar checks for data quality, regardless of whether their experiments are conducted online or in the lab.

Conclusion

A conservative interpretation of our findings suggests that (a) suppression of negative emotion slightly worsens verbal memory on a secondary task relative to control, but not necessarily relative to reappraisal; and (b) neither suppression nor reappraisal has consistent effects on nonverbal memory on a focal task relative to control. The bounded nature of these findings is notable given that all experiments successfully manipulated emotion regulation and were powered above 80% to detect small effects. This work reinforces the importance of replicability in the emotion literature, requiring the field to rigorously test some widely accepted assumptions. For now, it seems fair to say that suppression of negative emotion does seem to have a small effect of worsening verbal memory on a nonfocal task relative to control, but the jury is still out on whether—and if so, how—reappraisal impacts on memory. We urge the field to make a united effort to engage in systematic collaborative research that puts this question to bed once and for all.

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